

Lab Task 3

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Part A and B:

1. UDP (Fig in 1)

```
> Frame 15358: 72 bytes on wire (576 bits), 72 bytes captured (576
> Ethernet II, Src: 24:6a:0e:81:79:73 (24:6a:0e:81:79:73), Dst: Ro
> Internet Protocol Version 4, Src: 10.100.16.61, Dst: 8.8.8.8
> User Datagram Protocol, Src Port: 51339, Dst Port: 53
  ▾ Domain Name System (query)
    Transaction ID: 0xb280
    > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    ▾ Queries
      ▾ www.ietf.org: type A, class IN
        Name: www.ietf.org
        [Name Length: 12]
        [Label Count: 3]
        Type: A (Host Address) (1)
        Class: IN (0x0001)
        [Response In: 15739]
```

2. Destination Port: 53 (Fig in 1)

, Source Port: 53 (Fig in 5)

3. Query DNS Address: 8.8.8.8 , Local DNS Address: 8.8.8.8, 8.8.4.4, Yes (Fig in 1)

```
DNS Servers . . . . . : 8.8.8.8
                        8.8.4.4
```

4. Type: A, No answers (Fig in 1)

5. Two Answers. Answers showing -> Name, Type, Class, TTL, Data length, Address (Fig in 5)

```

> Frame 15/39: 104 bytes on wire (832 bits), 104 bytes captured (832
> Ethernet II, Src: Routerbo_a0:7d:57 (48:8f:5a:a0:7d:57), Dst: 24:
> Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.100.16.61
> User Datagram Protocol, Src Port: 53, Dst Port: 51339
▼ Domain Name System (response)
  Transaction ID: 0xb280
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  > Queries
  ▼ Answers
    ▼ www.ietf.org: type A, class IN, addr 104.16.44.99
      Name: www.ietf.org
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 3 (3 seconds)
      Data length: 4
      Address: 104.16.44.99
    ▼ www.ietf.org: type A, class IN, addr 104.16.45.99
      Name: www.ietf.org
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 3 (3 seconds)
      Data length: 4
      Address: 104.16.45.99
    [Request In: 15358]
    [Time: 0.060100000 seconds]

```

6. Destination Port: 53 (Fig in 6), Source Port: 53 (Fig in 9)

```

C:\Users\610.02.IP61>ipconfig/flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\610.02.IP61>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
DNS request timed out.
    timeout was 2 seconds.
Name:     google.com
Address:  142.251.223.238

```

```

> Frame 11617: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0
> Ethernet II, Src: 24:6a:0e:81:79:73 (24:6a:0e:81:79:73), Dst: 08:00:2b:01:02:03
> Internet Protocol Version 4, Src: 10.100.16.61, Dst: 8.8.8.8
> User Datagram Protocol, Src Port: 57478, Dst Port: 53
▼ Domain Name System (query)
    Transaction ID: 0x0002
    > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
▼ Queries
    ▼ google.com: type A, class IN
        Name: google.com
        [Name Length: 10]
        [Label Count: 2]
        Type: A (Host Address) (1)
        Class: IN (0x0001)
\[Response In: 11619\]

```

7. Query DNS Address: 8.8.8.8 (Fig in 1), Local DNS Address: 8.8.8.8, 8.8.4.4, Yes same

```

DNS Servers . . . . . : 8.8.8.8
                        8.8.4.4

```

8. Type: A, No Answers (Fig in 6)
9. 1 Answers (Fig in 9)

```

> Frame 11617: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0
> Ethernet II, Src: 24:6a:0e:81:79:73 (24:6a:0e:81:79:73), Dst: 08:00:2b:01:02:03
> Internet Protocol Version 4, Src: 10.100.16.61, Dst: 8.8.8.8
> User Datagram Protocol, Src Port: 57478, Dst Port: 53
▼ Domain Name System (query)
  Transaction ID: 0x0002
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
▼ Queries
  ▼ google.com: type A, class IN
    Name: google.com
    [Name Length: 10]
    [Label Count: 2]
    Type: A (Host Address) (1)
    Class: IN (0x0001)
    [Response In: 11619]

```

Task: Wireshark Lab: IP

1. IP Address: 192.168.1.102

Time	Source	Destination	Protocol
8 6.163045	192.168.1.102	128.59.23.100	ICMP
9 6.176826	10.216.228.1	192.168.1.102	ICMP
10 6.188629	192.168.1.102	128.59.23.100	ICMP
11 6.202957	24.218.0.153	192.168.1.102	ICMP

2. Protocol:

```

> Time to Live: 1
  Protocol: ICMP (1)
  Header Checksum: 0x2d2c

```

3. 20bytes

```

> Frame 8: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
> Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: L: 08:00:2b:01:02:03
▼ Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.59.23.100
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x32d0 (13008)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0

```

Payload = 84(total data) - 20 (header)

4. No.

```
Identification: 0x32d0 (13008)
  000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
```

5. Changeable: Transaction ID/Identification, TTL, Checksum

6. Unchangeable: Version, Header length, Destination, Source

8.

```
  Ethernet II, Src: Actionte_8a:70:1a (00:20:e0
    > Destination: LinksysG_da:af:73 (00:06:25:d
    > Source: Actionte_8a:70:1a (00:20:e0:8a:70:
      Type: IPv4 (0x0800)
  Internet Protocol Version 4, Src: 192.168.1.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP:
      Total Length: 84
      Identification: 0x32d0 (13008)
    > 000. .... = Flags: 0x0
      ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 1
```

Fragmentation:

1.

```
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 568
Identification: 0x3333 (13107)
✓ 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0... .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0001 0111 0010 = Fragment Offset: 2960
✓ Time to Live: 4
    ✓ [Expert Info (Note/Sequence): "Time To Live" only 4]
        ["Time To Live" only 4]
        [Severity level: Note]
        [Group: Sequence]
    Protocol: ICMP (1)
    Header Checksum: 0x2673 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.102
    Destination Address: 128.59.23.100
✓ [3 IPv4 Fragments (3508 bytes): #281(1480), #282(1480), #283(548)]
    [Frame: 281, payload: 0-1479 (1480 bytes)]
    [Frame: 282, payload: 1480-2959 (1480 bytes)]
    [Frame: 283, payload: 2960-3507 (548 bytes)]
    [Fragment count: 3]
    [Reassembled IPv4 length: 3508]
    [Reassembled IPv4 data: 080098cc0300ae03383020aaaaaaaaaaaaaaaa]
```

347	53.626815	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3344)
348	53.627582	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=334)
349	53.628465	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48643/958, ttl=7 (no
350	53.657055	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3345)
351	53.657813	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=334)
352	53.658658	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48899/959, ttl=8 (no
353	53.676909	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3346)
354	53.677575	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=334)

> Frame 347: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)

> Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysG_da:af:73 (00:06:25:da:af:73)

▼ Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.59.23.100

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0000 00.. = Differentiated Services Codepoint: Default (0)

.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)

Total Length: 1500

Identification: 0x3344 (13124)

▼ 001. = Flags: 0x1, More fragments

0... = Reserved bit: Not set

.0.. = Don't fragment: Not set

..1. = More fragments: Set

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 7

Protocol: ICMP (1)

Header Checksum: 0x0130 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.102

Destination Address: 128.59.23.100

[\[Reassembled IPv4 in frame: 349\]](#)

> Data (1480 bytes)

2.

349	53.613079	24.210.0.133	192.168.1.102	ICMP	70	Time-to-live exceeded (time to live exceeded in transit)
347	53.626815	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3344) [R
348	53.627582	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3344)
349	53.628465	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48643/958, ttl=7 (no
350	53.657055	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3345) [R
351	53.657813	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3345)
352	53.658658	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48899/959, ttl=8 (no
353	53.676909	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3346) [R
354	53.677575	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3346)

> Frame 348: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)

> Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysG_da:af:73 (00:06:25:da:af:73)

▼ Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.59.23.100

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0000 00.. = Differentiated Services Codepoint: Default (0)

.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)

Total Length: 1500

Identification: 0x3344 (13124)

▼ 001. = Flags: 0x1, More fragments

0... = Reserved bit: Not set

.0.. = Don't fragment: Not set

..1. = More fragments: Set

...0 0000 1011 1001 = Fragment Offset: 1480

Time to Live: 7

Protocol: ICMP (1)

Header Checksum: 0x0077 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.102

Destination Address: 128.59.23.100

[\[Reassembled IPv4 in frame: 349\]](#)

> Data (1480 bytes)

347	53.626815	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3344) [
348	53.627582	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3344
349	53.628465	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48643/958, ttl=7 (n
350	53.657055	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3345) [
351	53.657813	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3345
352	53.658658	192.168.1.102	128.59.23.100	ICMP	582	Echo (ping) request id=0x0300, seq=48899/959, ttl=8 (n
353	53.676909	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=0, ID=3346) [
354	53.677575	192.168.1.102	128.59.23.100	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=1480, ID=3346

> Frame 349: 582 bytes on wire (4656 bits), 582 bytes captured (4656 bits)

> Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysG_da:af:73 (00:06:25:da:af:73)

Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.59.23.100

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

✓ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0000 00.. = Differentiated Services Codepoint: Default (0)

.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)

Total Length: 568

Identification: 0x3344 (13124)

✓ 000. = Flags: 0x0

0... = Reserved bit: Not set

.0.. = Don't fragment: Not set

..0. = More fragments: Not set

...0 0001 0111 0010 = Fragment Offset: 2960

Time to Live: 7

Protocol: ICMP (1)

Header Checksum: 0x2362 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.102

Destination Address: 128.59.23.100

✓ [3 IPv4 Fragments (3508 bytes): #347(1480), #348(1480), #349(548)]

[Frame: 347, payload: 0-1479 (1480 bytes)]

[Frame: 348, payload: 1480-2959 (1480 bytes)]

[Frame: 349, payload: 2960-3507 (548 bytes)]

[Fragment count: 3]

[Reassembled IPv4 length: 3508]

-----11-b-d IPv4 data: 00000000-00000000-00000000-00000000----- 1

3. 3 fragments (Fig in 2)
4. Header checksum, Flag, Fragment Offset, frame Number, Total length

Task 3: Wireshark Lab: TCP

1.